MATHS MANTHAN

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Subject: Mathematics **CBSE CLASS 10 – WEEKLY CHAPTER TEST – 05** **Time:** 45 Minutes

Class: Class X (CBSE) **Chapter 4: Quadratic Equations** **MM: 20**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

Section A: MCQ & Assertion-Reason Questions

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Q1. If $(a^2+b^2)x^2 - 2(ac+bd)x + (c^2+d^2) = 0$ has equal roots, then: [1]

(A) $ab = cd$ **(B)** $ad = bc$

(C) $ad = \sqrt{bc}$ **(D)** $ab = \sqrt{cd}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $ad = bc$

Comparing $(a^2+b^2)x^2 - 2(ac+bd)x + (c^2+d^2) = 0$ with $Ax^2 + Bx + C = 0$, we have $A = a^2+b^2$, $B = -2(ac+bd)$, and $C = c^2+d^2$.

Formula Used: Equal real roots $\iff D = B^2 - 4AC = 0$

Setting $D = 0$:

$$[-2(ac+bd)]^2 - 4(a^2+b^2)(c^2+d^2) = 0$$
$$4(a^2c^2 + 2abcd + b^2d^2) - 4(a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2) = 0$$
$$-4(a^2d^2 - 2abcd + b^2c^2) = 0$$
$$(ad - bc)^2 = 0 \implies \boxed{ad = bc}$$

Hence, option **(B)** is correct. **[1 mark]**

Q2. The roots of the quadratic equation $x^2 + 3x - 10 = 0$ are: [1]

(A) $2,\ -5$ **(B)** $-2,\ 5$

(C) $2,\ 5$ **(D)** $-2,\ -5$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $2,\ -5$

Given equation: $x^2 + 3x - 10 = 0$. We find two numbers whose sum is 3 and product is -10, namely $+5$ and -2.

Formula Used: $x^2 + (\alpha+\beta)x + \alpha\beta = (x+\alpha)(x+\beta) = 0$

Factorising:

$$x^2 + 5x - 2x - 10 = 0 \implies x(x+5) - 2(x+5) = 0 \implies (x-2)(x+5) = 0 \implies \boxed{x = 2,\ -5}$$

Therefore, option **(A)** is correct. **[1 mark]**

Q3. Which of the following quadratic equations has the sum of its roots as 4? [1]

(A) $2x^2 - 4x + 8 = 0$ **(B)** $-x^2 + 4x + 4 = 0$

(C) $\sqrt{2}x^2 - \frac{4}{\sqrt{2}}x + 1 = 0$ **(D)** $4x^2 - 4x + 4 = 0$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: **(B)** $-x^2 + 4x + 4 = 0$

For a quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$, the sum of roots is given by:

Formula Used: Sum of roots $= -\frac{b}{a}$

Computing $-\frac{b}{a}$ for each option:

(A) $-\frac{-4}{2} = 2 \neq 4$

(B) $-\frac{4}{-1} = 4$ (Matches the required sum!)

(C) $-\frac{-4/\sqrt{2}}{\sqrt{2}} = \frac{4}{2} = 2 \neq 4$

(D) $-\frac{-4}{4} = 1 \neq 4$

Hence, option **(B)** is correct. **[1 mark]**

Q4. The quadratic equation $x^2 + x + 1 = 0$ has: [1]

(A) real and equal roots **(B)** irrational roots

(C) real and distinct roots **(D)** no real roots

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: **(D) no real roots**

Comparing $x^2 + x + 1 = 0$ with $ax^2 + bx + c = 0$, we have $a = 1$, $b = 1$, and $c = 1$.

Formula Used: $D = b^2 - 4ac < 0 \implies$ No real roots

Calculating D:

$$D = (1)^2 - 4(1)(1) = 1 - 4 = -3 < 0$$

Since $D < 0$, the equation has no real roots. Thus, option **(D)** is correct. **[1 mark]**

Q5. **Assertion-Reason:** [1]

Assertion (A): If a and c are of opposite signs, then $ax^2 + bx + c = 0$ has real and distinct roots.

Reason (R): If a and c have opposite signs, $ac < 0 \implies -4ac > 0$, so $D = b^2 - 4ac > 0$.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true, but R is not the correct explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: **(A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).**

Because a and c have opposite signs, $a \neq 0$, $c \neq 0$, and $ac < 0$, which gives $-4ac > 0$.

Formula Used: $b^2 \geq 0$ and $-4ac > 0 \implies D = b^2 - 4ac > 0$

Since the discriminant D is strictly positive, $ax^2 + bx + c = 0$ necessarily possesses two real and distinct roots.

Thus Assertion (A) is **true**, Reason (R) is **true**, and Reason (R) correctly explains Assertion (A). **[1 mark]**

Q6. If one root of the quadratic equation $2x^2 + kx - 6 = 0$ is 2, find the value of k and the other root. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let $p(x) = 2x^2 + kx - 6$. Since $x = 2$ is a root of $p(x) = 0$, we must have $p(2) = 0$:

Formula Used: If α is a root of $ax^2 + bx + c = 0$, then $a\alpha^2 + b\alpha + c = 0$ and $\alpha\beta = \dfrac{c}{a}$

Substituting $x = 2$:

$$2(2)^2 + k(2) - 6 = 0 \implies 8 + 2k - 6 = 0 \implies 2k + 2 = 0 \implies k = -1 \quad \textbf{[1 mark]}$$

Substituting $k = -1$ into $2x^2 + kx - 6 = 0$ yields $2x^2 - x - 6 = 0$.
Factorising $(2 \times (-6) = -12;\ -4 + 3 = -1)$:

$$2x^2 - 4x + 3x - 6 = 0 \implies 2x(x-2) + 3(x-2) = 0 \implies (x-2)(2x+3) = 0$$

Therefore, $x = 2$ or $x = -\dfrac{3}{2}$.

Hence, $\boxed{k = -1 \text{ and the other root is } -\dfrac{3}{2}}$. **[1 mark]**

Q7. Solve the following equation for x: $\dfrac{x-1}{x-2} + \dfrac{x-3}{x-4} = 3\dfrac{1}{3}, \quad x \neq 2,\ 4.$ [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Notice $3\dfrac{1}{3} = \dfrac{10}{3}$. Combining the fractions on the LHS over the common denominator $(x-2)(x-4)$:

Formula Used: $\dfrac{x-1}{x-2} + \dfrac{x-3}{x-4} = \dfrac{(x-1)(x-4) + (x-3)(x-2)}{(x-2)(x-4)}$

$$\frac{(x^2 - 5x + 4) + (x^2 - 5x + 6)}{x^2 - 6x + 8} = \frac{10}{3}$$

$$\frac{2x^2 - 10x + 10}{x^2 - 6x + 8} = \frac{10}{3} \quad \textbf{[1 mark]}$$

Dividing both sides by 2:

$$\frac{x^2 - 5x + 5}{x^2 - 6x + 8} = \frac{5}{3} \implies 3(x^2 - 5x + 5) = 5(x^2 - 6x + 8) \implies 3x^2 - 15x + 15 = 5x^2 - 30x + 40 \quad \textbf{[1 mark]}$$

Rearranging into standard form:

$$2x^2 - 15x + 25 = 0 \implies 2x^2 - 10x - 5x + 25 = 0 \implies 2x(x-5) - 5(x-5) = 0 \implies (x-5)(2x-5) = 0$$

This gives $x = 5$ or $x = \dfrac{5}{2}$, both satisfying the domain restrictions $x \neq 2, 4$.

Hence, the solutions are $\boxed{x = 5 \text{ and } x = \dfrac{5}{2}}$. **[1 mark]**

Q8. Solve for x by factorization: $36x^2 - 12ax + (a^2 - b^2) = 0$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given quadratic equation: $36x^2 - 12ax + (a^2 - b^2) = 0$.
Using the algebraic identity $a^2 - b^2 = (a - b)(a + b)$:

Formula Used: $A^2 - B^2 = (A - B)(A + B)$

We factorise the constant term as $(a - b)(a + b)$. Notice that:

$$-6(a-b)-6(a+b) = -6a+6b-6a-6b = -12a \quad \text{and} \quad [-6(a-b)]\,[-6(a+b)] = 36(a^2-b^2) \quad \textbf{[1 mark]}$$

Splitting the middle term:

$$\begin{aligned} 36x^2 - 6(a-b)x - 6(a+b)x + (a-b)(a+b) &= 0 \\ 6x\big[6x - (a-b)\big] - (a+b)\big[6x - (a-b)\big] &= 0 \quad \textbf{[1 mark]} \\ \big[6x - (a-b)\big]\big[6x - (a+b)\big] &= 0 \end{aligned}$$

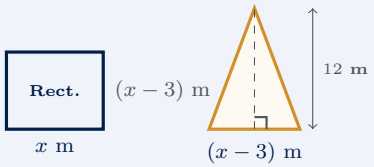
Equating each linear factor to zero:

$$6x = a - b \implies x = \frac{a-b}{6} \qquad \text{or} \qquad 6x = a + b \implies x = \frac{a+b}{6}$$

Hence, the roots are $\boxed{x = \dfrac{a-b}{6} \text{ and } x = \dfrac{a+b}{6}}$. **[1 mark]**

Q9. A rectangular park is to be designed whose breadth is 3 m less than its length. Its area is to be 4 m^2 more than the area of a park in the shape of an isosceles triangle with base as the breadth of the rectangular park and altitude 12 m. Find the length and breadth of the rectangular park. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Let the length of the rectangular park be x m ($x > 3$). Then its breadth is $(x - 3)$ m. The isosceles triangular park has its base equal to $(x - 3)$ m and altitude 12 m, as illustrated in the side-by-side layout diagram.

Formula Used: Area of rectangle $= l \times b$, Area of triangle $= \dfrac{1}{2} \times \text{base} \times \text{altitude}$

Writing areas in terms of x:

- Area of rectangular park $= x(x - 3)$ m^2.
- Area of isosceles triangular park $= \dfrac{1}{2} \times (x - 3) \times 12 = 6(x - 3)$ m^2. **[1 mark]**

According to the given condition:

$$\begin{aligned} x(x-3) &= 6(x-3) + 4 \\ x^2 - 3x &= 6x - 18 + 4 \implies x^2 - 9x + 14 = 0 \quad \textbf{[1 mark]} \end{aligned}$$

Factorising ($-7 \times (-2) = 14$ and $-7 - 2 = -9$):

$$x^2 - 7x - 2x + 14 = 0 \implies (x-7)(x-2) = 0 \implies x = 7 \text{ or } x = 2$$

If $x = 2$, then breadth $= x - 3 = -1$ m < 0, which is impossible; hence $x = 2$ is rejected.
Thus, length $= x = 7$ m and breadth $= 7 - 3 = 4$ m. (Check: $28 = 4 + \frac{1}{2}(4)(12)$.)
Therefore, $\boxed{\text{Length} = 7 \text{ m and Breadth} = 4 \text{ m}}$. **[1 mark]**

Q10. Case Study: Agricultural Farm Parceling & Land Allocation

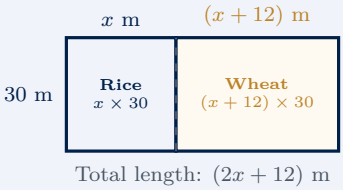
A progressive farmer owns a rectangular agricultural plot of total area 1260 m^2. He divides the land into two rectangular segments: a rice cultivation plot of dimensions x m$\times$30 m, and a wheat cultivation plot of dimensions $(x + 12)$ m $\times$ 30 m, such that the combined farm has a uniform width of 30 m and total length $(2x + 12)$ m. [4]

(i) Formulate the algebraic equation representing the total area of the farm in terms of x. **[1]**

(ii) Solve the equation to find the value of x. **[1]**

(iii) Calculate the separate areas of the wheat field and the rice field. **OR** Find the total cost of fencing the entire outer boundary of the 1260 m^2 farm at Rs. 40 per metre. **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



As shown in the partitioned farm layout alongside, the entire rectangle of area 1260 m^2 has uniform width 30 m and is split into a rice plot of length x m and an adjacent wheat plot of length $(x + 12)$ m, giving total length $(2x + 12)$ m.

We know the area and perimeter formulas for a rectangle:

Formula Used: Area = Length $\times$ Width, Perimeter = 2(Length + Width)

(i) Algebraic Equation for Total Area: Since combined length = $(2x+12)$ m and uniform width = 30 m:

$$\text{Total Area} = 30(2x + 12) = 1260 \implies \boxed{30(2x + 12) = 1260} \quad (\text{or } 60x + 360 = 1260)$$

Simplifying further gives $2x + 12 = 42$. **[1 mark]**

(ii) Solving for x: Dividing $30(2x + 12) = 1260$ by 30:

$$2x + 12 = \frac{1260}{30} = 42 \implies 2x = 42 - 12 = 30 \implies \boxed{x = 15 \text{ m}} \quad \textbf{[1 mark]}$$

(iii) First Part (Separate Areas): Substituting $x = 15$ m:

- Rice field length = 15 m, width = 30 m $\implies$ Area of rice field = $15 \times 30 = 450$ m^2.
- Wheat field length = $15 + 12 = 27$ m, width = 30 m $\implies$ Area of wheat field = $27 \times 30 = 810$ m^2.

Verification: $450 + 810 = 1260$ m^2. Hence, $\boxed{\text{Wheat} = 810 \text{ m}^2,\ \text{Rice} = 450 \text{ m}^2}$. **[2 marks]**

OR (Alternative Part — Fencing Cost): Total farm length = $2(15) + 12 = 42$ m and width = 30 m.
Outer perimeter = 2(Length + Width) = $2(42 + 30) = 2(72) = 144$ m.
Total cost of fencing at Rs. 40 per metre = 144×40 = Rs. 5760.
Hence, the total fencing cost is $\boxed{\text{Rs. } 5760}$. **[2 marks]**

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